

SCHEME OF VALUATION

Course: Engineering Mathematics-II

Examination: MID-I Examination

Max. Marks: 30

Academic Year: 2025-26

PART – A ($3 \times 2 = 6$ Marks)

1(a) Show that A is unitary if and only if $A^{-1} = A^*$.

- Definition of unitary matrix – 1 Mark
- Proof/verification – 1 Mark

Total: 2 Marks

1(b) Find the rank of the given matrix.

- Correct reduction process – 1 Mark
- Correct rank obtained – 1 Mark

Total: 2 Marks

1(c) Find the eigenvalues of the given matrix.

- Formation of characteristic equation – 1 Mark
- Correct eigenvalues – 1 Mark

Total: 2 Marks

PART – B ($2 \times 12 = 24$ Marks)

2(a) Find the rank of matrix A by reducing it to Echelon Form.

- Row operations and reduction – 4 Marks
- Echelon form obtained – 1 Mark
- Correct rank – 1 Mark

Total: 6 Marks

2(b) Find the rank of matrix A by reducing it to Normal Form.

- Reduction process – 4 Marks
- Normal form obtained – 1 Mark
- Correct rank – 1 Mark

Total: 6 Marks

OR

3(a) Show that the given system is consistent and solve it.

- Formation of augmented matrix – 2 Marks
- Consistency verification – 2 Marks
- Correct solution – 2 Marks

Total: 6 Marks

3(b) Solve the equations using Gauss Elimination Method.

- Formation of augmented matrix – 2 Marks
- Elimination process – 2 Marks
- Correct solution – 2 Marks

Total: 6 Marks

4(a) Find the eigenvalues and corresponding eigenvectors.

- Characteristic equation – 2 Marks
- Eigenvalues – 2 Marks
- Eigenvectors – 2 Marks

Total: 6 Marks

4(b) Verify Cayley–Hamilton Theorem and find A^{-1} .

- Characteristic equation – 2 Marks
- Verification of theorem – 2 Marks
- Computation of A^{-1} – 2 Marks

Total: 6 Marks

OR

5(a) Determine the modal matrix P and verify diagonalization.

- Eigenvalues and eigenvectors – 2 Marks
- Modal matrix formation – 2 Marks
- Verification of diagonal matrix – 2 Marks

Total: 6 Marks

5(b) Reduce the quadratic form to canonical form by orthogonal transformation.

- Matrix representation – 2 Marks
- Orthogonal transformation – 2 Marks
- Canonical form obtained – 2 Marks

Total: 6 Marks